

Fourier Transform - Table

x Domain	k Domain
$f(x) = \mathcal{F}^{-1}\{F(k)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k)e^{ikx} dk$	$F(k) = \mathcal{F}\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)e^{-ikx} dx$
1	$\sqrt{2\pi}\delta(k)$
$\delta(x)$	$\frac{1}{\sqrt{2\pi}}$
$\delta(x - c)$	$\frac{1}{\sqrt{2\pi}}e^{-ikc}$
$u(x)$	$\frac{1}{\sqrt{2\pi}}(\pi\delta(k) + \frac{1}{ik})$
$-0.5 + u(x)$	$\frac{1}{\sqrt{2\pi}}\frac{1}{ik}$
$e^{-bx}u(x), \quad (b > 0)$	$\frac{1}{\sqrt{2\pi}}\frac{1}{ik+b}$
$\cos k_0x$	$\sqrt{\frac{\pi}{2}}[\delta(k + k_0) + \delta(k - k_0)]$
$\cos(k_0x + \theta)$	$\sqrt{\frac{\pi}{2}}[e^{-i\theta}\delta(k + k_0) + e^{i\theta}\delta(k - k_0)]$
$\sin k_0x$	$i\sqrt{\frac{\pi}{2}}[\delta(k + k_0) - \delta(k - k_0)]$
$\sin(k_0x + \theta)$	$i\sqrt{\frac{\pi}{2}}[e^{-i\theta}\delta(k + k_0) - e^{i\theta}\delta(k - k_0)]$
$\text{rect}(ax)$	$\frac{1}{\sqrt{2\pi a^2}}\text{sinc}\left(\frac{k}{2\pi a}\right)$
$\text{sinc}(ax)$	$\frac{1}{\sqrt{2\pi a^2}}\text{rect}\left(\frac{k}{2\pi a}\right)$
$\text{tri}(ax) = (1 - a x)\text{rect}\left(\frac{ax}{2}\right)$	$\frac{1}{\sqrt{2\pi a^2}}\text{sinc}^2\left(\frac{k}{2\pi a}\right)$
$\text{sinc}^2(ax)$	$\frac{1}{\sqrt{2\pi a^2}}\text{tri}\left(\frac{k}{2\pi a}\right)$
$e^{-a x }, \quad \Re\{a\} > 0$	$\sqrt{\frac{\pi}{2}}\frac{a}{a^2 + k^2}$
$e^{-ax^2}, \quad \Re\{a\} > 0$	$\frac{1}{\sqrt{2a}}e^{-\frac{k^2}{4a}}$
$\cos(ax^2)$	$\frac{1}{\sqrt{2a}}\cos\left(\frac{k^2}{4a} - \frac{\pi}{4}\right)$
$\sin(ax^2)$	$\frac{-1}{\sqrt{2a}}\sin\left(\frac{k^2}{4a} - \frac{\pi}{4}\right)$
$\frac{1}{\sqrt{ x }}$	$\frac{1}{\sqrt{ k }}$

Notes

1. $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$
2. $\text{rect}\left(\frac{x}{\tau}\right)$ is the rectangular pulse function of width τ
3. $\text{tri}\left(\frac{x}{\tau}\right)$ is the triangular pulse function of width 2τ
4. $u(x)$ is the Heaviside step function
5. $\delta(x)$ is the Dirac delta function